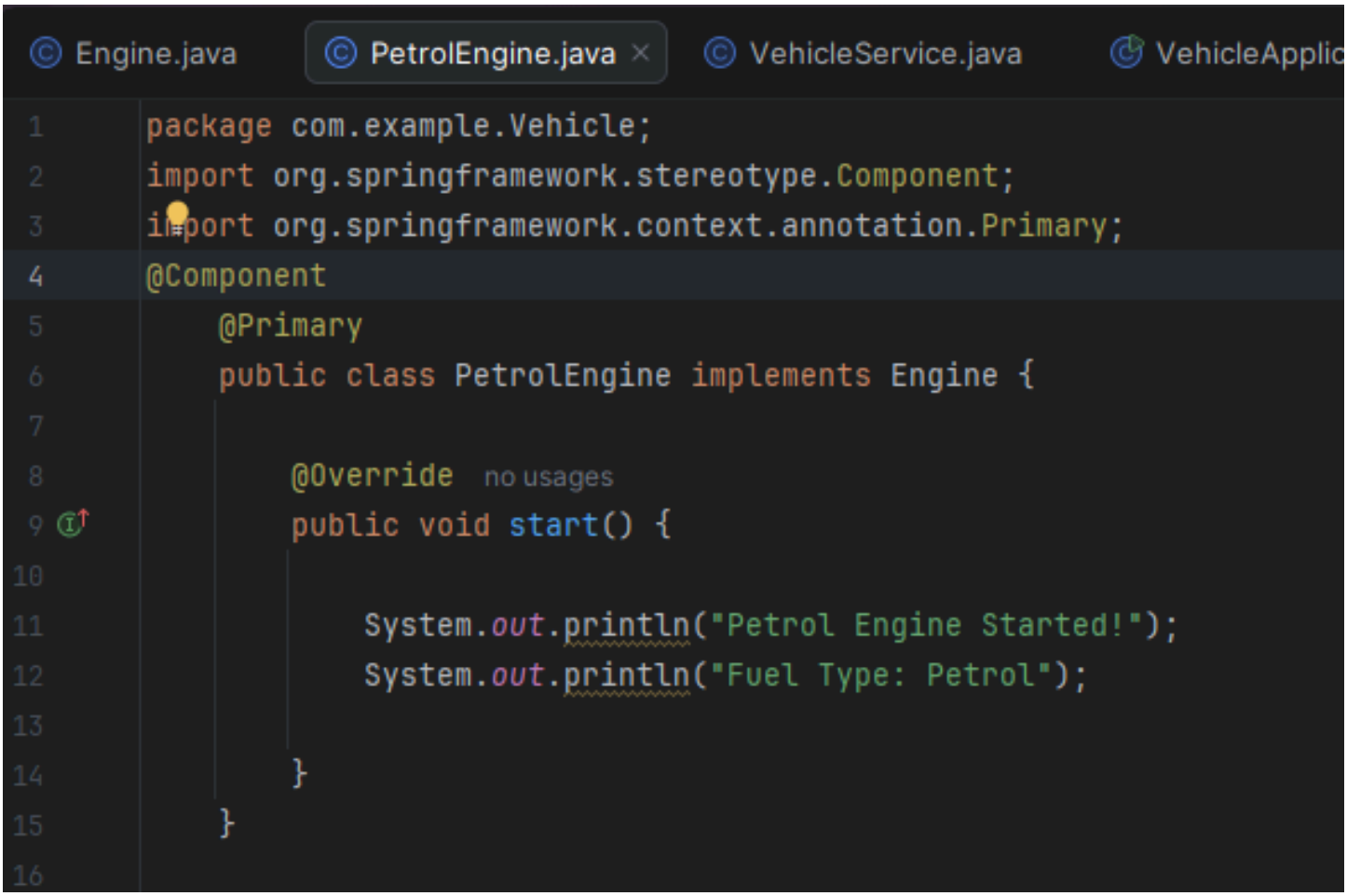
SPRINGBOOT

Loose Coupling

Using @Primary



Engine.java | PetrolEngine.java | VehicleService.java | VehicleApplication.java

```java
package com.example.Vehicle;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;
    @Component
    public class VehicleService {
        private final Engine engine;  2 usages

        @Autowired
        public VehicleService(Engine engine) {

            this.engine = engine;

        }
        public void drive() {  no usages

            engine.start();

            System.out.println("Vehicle is Ready to Drive!");

        }
    }

```

Engine.java | PetrolEngine.java | VehicleService.java | VehicleApplication.java

```java
package com.example.Vehicle;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.boot.CommandLineRunner;

@SpringBootApplication
public class VehicleApplication
        implements CommandLineRunner {

    private final VehicleService vehicleService;  2 usages

    public VehicleApplication(  no usages
            VehicleService vehicleService) {

        this.vehicleService = vehicleService;

    }

    public static void main(String[] args) {

        SpringApplication.run(
                VehicleApplication.class, args);

    }

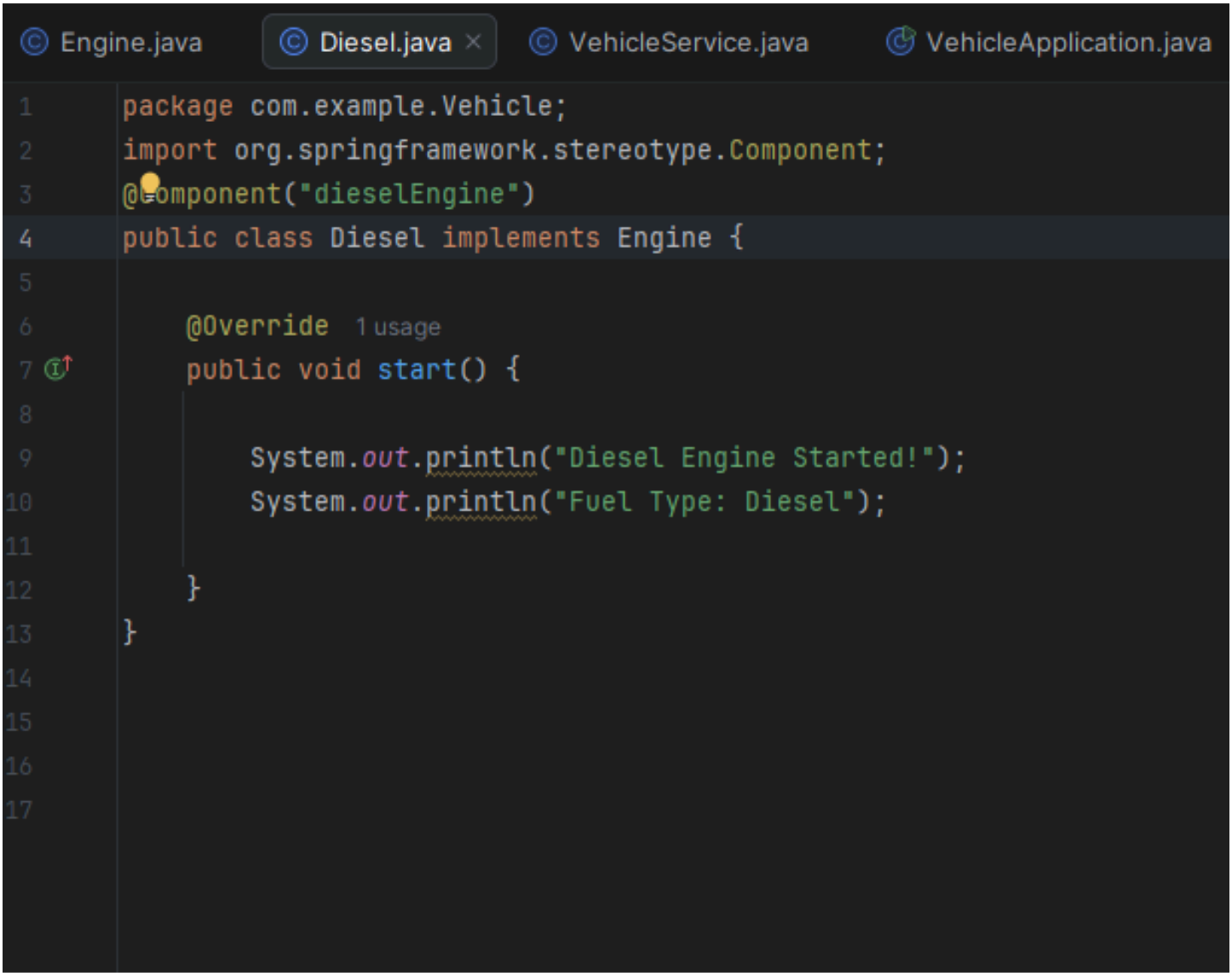
    @Override
    public void run(String... args) {
```

```
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 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
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 :: Spring Boot ::                (v4.1.2)

2026-11-06T22:24:45.690+05:30  INFO 19872 --- [Vehicle] [  restartedMain] com.example.Vehicle.VehicleApplication   : Starting VehicleApplication using Ja
2026-11-06T22:24:45.694+05:30  INFO 19872 --- [Vehicle] [  restartedMain] com.example.Vehicle.VehicleApplication   : No active profile set, falling back
2026-11-06T22:24:45.859+05:30  INFO 19872 --- [Vehicle] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! S
2026-11-06T22:24:45.859+05:30  INFO 19872 --- [Vehicle] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging c
2026-11-06T22:24:47.364+05:30  INFO 19872 --- [Vehicle] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat initialized with port 8080 (h
2026-11-06T22:24:47.389+05:30  INFO 19872 --- [Vehicle] [  restartedMain] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-11-06T22:24:47.390+05:30  INFO 19872 --- [Vehicle] [  restartedMain] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tom
2026-11-06T22:24:47.446+05:30  INFO 19872 --- [Vehicle] [  restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initiali
2026-11-06T22:24:47.963+05:30  INFO 19872 --- [Vehicle] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat started on port 8080 (http) w
2026-11-06T22:24:47.973+05:30  INFO 19872 --- [Vehicle] [  restartedMain] com.example.Vehicle.VehicleApplication   : Started VehicleApplication in 3.176
Petrol Engine Started!
Fuel Type: Petrol
Vehicle is Ready to Drive!
```

Using @qualifier



Engine.java | Diesel.java | VehicleService.java | VehicleApplication.j

```java
package com.example.Vehicle;
import org.springframework.stereotype.Component;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.beans.factory.annotation.Qualifier;
@Component
public class VehicleService {

    private final Engine engine;  2 usages
    @Autowired
    public VehicleService(
            @Qualifier("dieselEngine") Engine engine) {

        this.engine = engine;

    }

    public void drive() {  1 usage
        engine.start();
        System.out.println("Vehicle is Ready to Drive!");

    }
}
```

Engine.java | Diesel.java | VehicleService.java | VehicleApplication.java

```java
package com.example.Vehicle;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.boot.CommandLineRunner;
@SpringBootApplication
public class VehicleApplication implements CommandLineRunner {
    private final VehicleService vehicleService;  2 usages
    public VehicleApplication(VehicleService vehicleService) {  no usages
        this.vehicleService = vehicleService;
    }
    public static void main(String[] args) {

        SpringApplication.run(VehicleApplication.class, args);

    }
    @Override
    public void run(String... args) {

        vehicleService.drive();

    }
}
```

```
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( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
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 :: Spring Boot ::                (v4.1.2)

2026-11-06T22:50:02.797+05:30  INFO 21184 --- [Vehicle] [  restartedMain] com.example.Vehicle.VehicleApplication   : Starting VehicleApplication u
2026-11-06T22:50:02.802+05:30  INFO 21184 --- [Vehicle] [  restartedMain] com.example.Vehicle.VehicleApplication   : No active profile set, fallin
2026-11-06T22:50:02.932+05:30  INFO 21184 --- [Vehicle] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults ac
2026-11-06T22:50:02.933+05:30  INFO 21184 --- [Vehicle] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related lo
2026-11-06T22:50:04.368+05:30  INFO 21184 --- [Vehicle] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat initialized with port
2026-11-06T22:50:04.391+05:30  INFO 21184 --- [Vehicle] [  restartedMain] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-11-06T22:50:04.392+05:30  INFO 21184 --- [Vehicle] [  restartedMain] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apa
2026-11-06T22:50:04.459+05:30  INFO 21184 --- [Vehicle] [  restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: i
2026-11-06T22:50:05.041+05:30  INFO 21184 --- [Vehicle] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat started on port 8080 (
2026-11-06T22:50:05.047+05:30  INFO 21184 --- [Vehicle] [  restartedMain] com.example.Vehicle.VehicleApplication   : Started VehicleApplication in
Diesel Engine Started!
Fuel Type: Diesel
Vehicle is Ready to Drive!
```